AI-Based Fraud Detection in Government Schemes

Submitted by:

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GitHub Repository Link: https://github.com/saniyashariff61281-sys/-fraud-detection-project.git

Introduction

Government welfare schemes such as subsidies, scholarships, and Direct Benefit Transfers (DBT) are essential for supporting citizens. However, these systems often face issues like fraudulent beneficiaries, duplicate identities, and abnormal transaction activities. Such problems lead to financial losses and reduced transparency. Therefore, there is a need for an intelligent system to detect fraud efficiently.

Problem Statement

Manual monitoring of large-scale beneficiary data is inefficient and error-prone. Fraudulent users may exploit loopholes by making abnormal transactions or creating fake identities. The problem is to design a system that can automatically detect such suspicious activities in government schemes.

Existing System

Current systems rely mostly on manual verification and rule-based checks. These approaches:

- Are time-consuming
- Cannot handle large datasets efficiently
- Fail to detect hidden patterns in data

Hence, they are not sufficient for modern large-scale fraud detection.

System Design

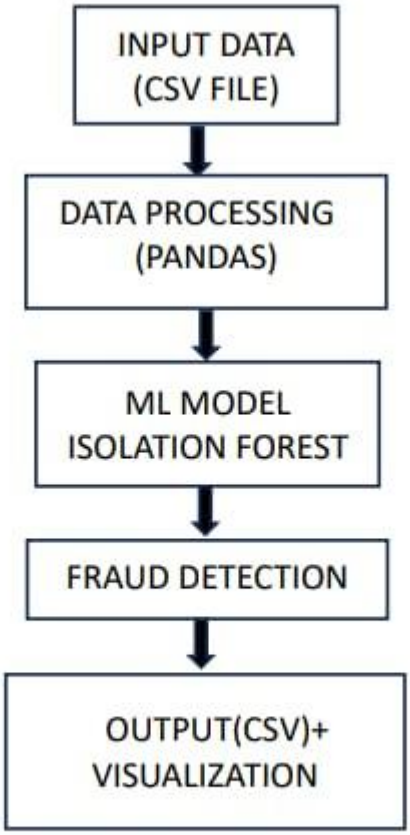
Architecture Flow

Data Input → Data Processing → Machine Learning Model → Fraud Detection → Output

Working

- Data is collected in CSV format

- Processed using Pandas
- Passed to the Isolation Forest model
- Model detects anomalies
- Results stored in output.csv



Tools & Technologies

- Python
- Pandas
- Scikit-learn
- VS Code

Objectives

- Detect fraudulent beneficiaries
- Identify abnormal transaction patterns
- Improve transparency

- Reduce financial losses
- Automate fraud detection using AI

Output

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS C:\Users\Dsf20\Downloads\New folder\fraud_project> pip install pandas scikit-learn
Requirement already satisfied: joblib>=1.3.0 in c:\users\dsf20\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\dsf20\appdata\local\programs\python\python313\lib\site-packages (from scikit-learn)
(3.6.0)
Requirement already satisfied: six>=1.5 in c:\users\dsf20\appdata\local\programs\python\python313\lib\site-packages (from python-dateutil>=2.8.2->pa
ndas) (1.17.0)

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Dsf20\Downloads\New folder\fraud_project> python fraud_detection.py
Fraud Detection Completed!
   user_id  amount  transactions  age  fraud
0        1    5000             2   25      0
1        2  200000            20   45      0
2        3    3000             1   22      0
3        4  150000            25   40      0
4        5    4000             2   23      0
5        6  250000            30   50      1
6        7    3500             1   21      0
7        8  100000            15   38      0
PS C:\Users\Dsf20\Downloads\New folder\fraud_project>
x to restart.                                                   Ln 21, Col 12   Spaces: 4   UTF-8   CRLF   {} Python
```

```
EXPLORER
FRAUD_PR...
data.csv
fraud_detection.py
output.csv

fraud_detection.py   output.csv   data.csv

output.csv
1  user_id,amount,transactions,age,fraud
2  1,5000,2,25,0
3  2,200000,20,45,0
4  3,3000,1,22,0
5  4,150000,25,40,0
6  5,4000,2,23,0
7  6,250000,30,50,1
8  7,3500,1,21,0
9  8,100000,15,38,0
10

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

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6        7    3500             1   21      0
7        8  100000            15   38      0
PS C:\Users\Dsf20\Downloads\New folder\fraud_project>
OUTLINE
TIMELINE
```

Conclusion

We successfully developed a machine learning-based fraud detection system using the Isolation Forest algorithm. The system identifies suspicious users based on abnormal transaction patterns and helps improve efficiency and transparency in government schemes.

Future Work

- Integration with real-time government databases
- Use of advanced AI models
- Dashboard visualization using Streamlit
- Aadhaar-based duplicate detection